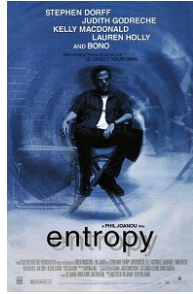


PHYSICS OF COMPLEX SYSTEMS

LECTURE AND TUTORIALS – PROF. DR. HAYE HINRICHSSEN – MAXIMILIAN SCHÄFFERER – SS 2026



EXERCISE 9.1: ENTROPY OF THE BINOMIAL DISTRIBUTION (3P)

The binomial distribution reads

$$P_n(k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

- (a) Use *Mathematica*[®] or similar software packages to compute the Shannon entropy (measured in *bit*) for $p = \frac{1}{2}$ and $n = 1, 2, 4, 8$. (1P)
- (b) Approximate the factorials by Stirlings formula, and Taylor-expand the resulting approximation in its maximum. Use this result to approximate the binomial distribution for $n \rightarrow \infty$ and $p = \frac{1}{2}$ by a normal distribution. (1P)
- (c) Compute the entropy of the approximated distribution, replacing the sum by a continuous integration. How does it grow with n ? (1P)

EXERCISE 9.2: GENERATING FUNCTION (3P)

A function which can be represented as

$$f(x) = \sum_{k=0}^n a_k x^k$$

is called the **generating function** of the coefficients a_k . In many cases, where the a_k are related to a certain combinatorical problem, $f(x)$ can be expressed in a closed form.

- (a) Determine the coefficients $\{a_k\}$ for $f(x) = (1+x)^n$ and for $f(x) = \frac{1}{1-bx}$. (1P)
- (b) Find the generating function for the coefficients starting with

$$\{a_k\} = \{0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \dots\}.$$

There is a nice website at oeis.org. Identify the series and search for “g.f.”, which stands for “generating function”. Verify your result e.g. with *Mathematica*[®]. (1P)

- (c) Let $f(x)$, $g(x)$, and $h(x) = f(x)g(x)$ be generating functions with the coefficients $\{a_k\}$, $\{b_k\}$, and $\{c_k\}$, respectively. Show that the c_k can be represented as a convolution of the $\{a_k\}$ and $\{b_k\}$. For simplicity ignore the radius of convergence. (1P)

EXERCISE 9.3: ENTROPY AND LAGRANGE MULTIPLIERS**(3P)**

Consider a Markov jump process, where each configuration $c \in \Omega$ is associated with a certain value $\chi_c \in \mathbb{R}$ (for example, the energy). Use the method of Lagrange multipliers to maximize the entropy

$$H = - \sum_{c \in \Omega} p_c \ln p_c$$

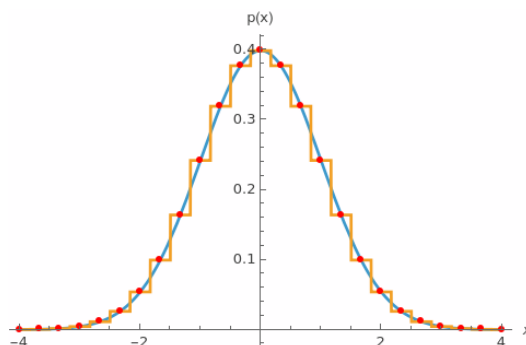
under the constraint that the average value $\bar{\chi} = \sum_c p_c \chi_c$ has a given positive value. Make sure that your result (the probability distribution p_c) is properly normalized. (3P)

EXERCISE 9.4: DIFFERENTIAL ENTROPY**(3P)**

Let $p(x)$ be a probability density normalized by $\int_{-\infty}^{+\infty} p(x) dx = 1$. Then the corresponding *differential entropy* is defined by

$$H_c = - \int_{-\infty}^{+\infty} p(x) \ln p(x) dx.$$

Consider a discretization of the x -axis into equidistant intervals of size Δx . Let $p_j = \frac{p(j \Delta x) \Delta x}{\mathcal{N}}$ be a discrete probability distribution which approximates $p(x)$, where $\mathcal{N} = \sum_{j=-\infty}^{+\infty} p(j \Delta x) \Delta x$ is the normalization.



- Show that in the limit $\Delta x \rightarrow 0$ the entropy H_c differs from the usual discrete Shannon entropy $H = - \sum_{j=-\infty}^{\infty} p_j \ln p_j$ only by an offset $\Delta H := H - H_c$. (1P)
- Compute this offset to leading order in Δx . (1P)
- Explain your findings qualitatively. (1P)

($\Sigma = 12P$)

Please submit your solution electronically as a single PDF file until Wednesday, June 17, 2026, via WueCampus.